

TABLE 33 ASIA PACIFIC: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Japan	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
India	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Rest of Asia Pacific	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Total	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 34 LATIN AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Rest of Latin America	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Total	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.3 WHOLE-GENOME SEQUENCING

7.3.1 ACCURATE REFERENCE GENOMES GENERATED FOR MICROBIAL IDENTIFICATION TO BOOST MARKET

Microbial whole-genome sequencing (WGS) has proved to be an important tool for mapping the genomes of novel organisms, finishing genomes of known organisms, or comparing genomes across multiple samples. This technology further helps researchers in generating accurate reference genomes for microbial identification and other comparative genomic studies. For functional studies of the microbiome, WGS metagenomic approaches are more commonly utilized. This is because WGS is affordable and particularly because of an array of computational tools to analyze microbiome sequencing data.

The advantages offered by this technique have resulted in increased adoption among scientists/researchers. As compared to other traditional genomic analysis methods, NGS-based microbial whole-genome sequencing eliminates labor-intensive cloning steps, saves time, and simplifies the workflow. This technique also helps to identify low-frequency variants and genome rearrangements that may be missed or are too expensive to identify using other methods.

TABLE 35 WHOLE-GENOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%
Europe	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%
Asia Pacific	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Latin America	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Middle East and Africa	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Total	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 36 NORTH AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	24.40	27.56	31.12	35.17	39.79	45.16	51.43	58.83	13.6%
Canada	2.53	2.83	3.16	3.54	3.97	4.46	5.03	5.70	12.5%
Total	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 37 EUROPE: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.17	3.61	4.11	4.68	5.33	6.10	7.00	8.08	14.5%
UK	2.84	3.24	3.71	4.24	4.85	5.57	6.41	7.42	14.9%
France	2.25	2.55	2.89	3.27	3.72	4.23	4.84	5.56	14.0%
Italy	0.63	0.72	0.81	0.91	1.03	1.17	1.33	1.53	13.5%
Spain	0.49	0.55	0.62	0.70	0.79	0.89	1.01	1.15	13.2%
Rest of Europe	5.32	5.98	6.72	7.55	8.50	9.59	10.86	12.35	12.9%
Total	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 38 ASIA PACIFIC: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Japan	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
India	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Rest of Asia Pacific	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Total	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 39 LATIN AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Rest of Latin America	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Total	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.4 SHOTGUN SEQUENCING

7.4.1 ADVANCEMENTS IN DATA ANALYSIS SOLUTIONS TO SUPPORT ADOPTION

Shotgun sequencing involves breaking up large DNA sequences into multiple small pieces/sequences and then reassembling them into larger contiguous pieces representing the original large DNA sequence. Shotgun sequencing is a relatively new approach used to analyze many organisms present in a given complex sample and comprehensively sequence all genes. This technique provides researchers or end users with the necessary information to evaluate community biodiversity and its related function in the ecosystem.

The advantages offered by shotgun metagenomic sequencing over capillary sequencing, PCR-based approaches, and traditional next-generation sequencing techniques, the growing adoption of shotgun metagenomic sequencing among end users, and the growing number of metagenomic sequencing research activities across the globe are the key factors driving the growth of the shotgun metagenomic sequencing market.